ECO372H1F — Assignment 1

Department of Economics, University of Toronto
Summer 2026
Due: May 31, 11:59 PM

Instructions

This assignment has two questions. Both ask you to write a short policy memo supported by empirical analysis. You will submit two files through Quercus:

1. A single PDF with both memos. Each memo is capped at **2 pages, single-spaced, 11pt font**. A cover page with your name and student number does not count toward the limit. Tables, figures, and any references are inside the 2-page limit. No appendices.

2. A single `.R` or `.Rmd` file that reproduces every number and figure in your written submission. The script should run end-to-end against the provided CSV files in the same directory.

A separate file, `R_syntax_reference.md`, is posted on Quercus alongside this assignment. It contains R syntax for various operations.

You may discuss the problems with classmates, but the writing and code you submit must be your own. You may use generative AI to debug code or clarify concepts. Any written text in your memos (interpretation, argument, recommendation) must be in your own words. AI-generated written content in your submission is treated as academic misconduct.

Plan for roughly 12–18 hours of work across the two questions over the two weeks. If you find yourself spending substantially more than that, please come to office hours.

Question 1 — Standardized Exams and University Performance

Scenario

You are a junior data analyst at the **British Columbia Ministry of Education**. The Deputy Minister is weighing a proposal to introduce province-wide standardized exams in BC for all core Grade 12 subjects (e.g. mathematics, the sciences, and English). The exams would count toward students' final grades and be administered under standardized conditions.

Supporters of the policy argue that standardized exams raise the rigour of high school instruction and prepare students better for university coursework. Critics argue they narrow the curriculum and create stress without improving outcomes.

Most Canadian provinces do not have comparable exams. In this scenario, for the sake of simplicity, let's assume that **Alberta expanded its mandatory diploma exam program in 2018** to cover additional core Grade 12 subjects, much like what BC is considering. **Ontario has never had province-wide standardized exams.**

The data

The Ministry has assembled an administrative dataset of students who graduated high school in Ontario, Alberta, or British Columbia between 2013 and 2022 and enrolled at a Canadian university.

The file `eco372_assignment1_q1_data.csv` contains:

Variable	**Description**
`student_id`	Unique student identifier
`province_hs`	Province where student attended high school (ON, AB, or BC)
`cohort_year`	Year the student graduated high school
`hs_gpa`	High school GPA on a 0–100 scale
`female`	1 if female, 0 otherwise
`parent_university`	1 if at least one parent has a university degree, 0 otherwise
`field`	University field of study (STEM, Humanities, Social Sciences, Business)
`university`	The Canadian university the student attended
`first_year_gpa`	First-year university GPA on a 0.0–4.0 scale

Assumptions you may make:

- Students attend universities across Canada. Assume university assignment is independent of the student's high school preparation — that is, students do not sort into universities based on their high school experience. This is a strong assumption but it lets you set aside one source of confounding for this exercise.
- The dataset is a random sample from the population of Ontario, Alberta, and BC high school graduates who attended university.
- There is no missing data and no measurement error.
- BC has never had a comparable standardized exam policy. BC data is included in the dataset for you to use as you see fit.

Your task

Write a memo to the Deputy Minister ($\leq$ 2 pages, single-spaced, 11pt). Assume they are roughly familiar with causal inference. Write in plain language and define any technical terms that aren't standard in an introductory causal inference course.

The memo must address:

1. Your estimate of the causal effect of Alberta's standardized exam policy on first-year university GPA, with a standard error (or confidence interval).
2. What this estimate means in plain language.
3. The research design behind your estimate: what method you used, why it is appropriate for this data, what assumption it relies on, and evidence (a figure, a formal test, or both) that the assumption is plausible here.
4. Threats to validity — what could still go wrong, and how worried you are about each.
5. A recommendation for BC. The Deputy Minister wants to know how confident they should be that what happened in Alberta tells them something about what would happen in BC. The dataset includes a third province — think about what that lets you say. Make the argument concrete; don't just claim BC is or isn't similar.

Tables and figures count toward the 2-page limit.

Question 2 — Insurance Premiums and Healthcare Engagement

Scenario

You are a junior data analyst at the **Ontario Ministry of Health**. The Ministry regulates a private health insurance market for working-age adults that supplements OHIP. Under current regulation, premiums rise sharply at **age 25**, when individuals are reclassified from a youth tier into an adult tier.

The Ministry is worried about a specific behavioural channel. High premiums at young ages may discourage people from engaging with the healthcare system during a formative period. Once someone stops going to the doctor regularly, that habit can stick — so a short-run drop in engagement at age 25 may translate into worse long-run health outcomes. The Ministry is considering raising the cutoff from 25 to 30 to delay the premium increase.

Before recommending the change, the Deputy Minister wants to know: does the age-25 cutoff actually cause a meaningful drop in healthcare utilization? If it doesn't, the policy change is unlikely to deliver the intended benefits.

The data

The file `eco372_assignment1_q2_data.csv` contains administrative data on individuals aged 20 to 30 who hold a private regulated health insurance policy in Ontario:

Variable	Description
`person_id`	Unique individual identifier
`age`	Age in years (continuous, with decimals)
`female`	1 if female, 0 otherwise
`income`	Annual income in CAD
`preexisting_condition`	1 if the individual has a documented pre-existing condition, 0 otherwise
`supplementary_insurance`	1 if the individual holds supplementary coverage in addition to the primary regulated policy, 0 otherwise
`annual_spending`	Total annual healthcare spending in CAD

Background you may use:

- Every individual in the dataset holds the primary regulated policy. Premiums under this policy rise sharply at age 25.
- A small share of individuals also hold a supplementary policy from a different insurer. The supplementary policy has its own age-based eligibility rules.
- The exact location of the age-25 cutoff is not publicly known; the regulation specifies it in a technical document that the general public does not read. You may therefore assume individuals are not able to time their insurance decisions to fall on a particular side of the cutoff.
- The dataset is a random sample from the population of working-age Ontarians holding this type

of private insurance. There is no missing data and no measurement error.

Your task

Similar to that of Question 1, please write a memo to the Deputy Minister (≤ 2 pages, single-spaced, 11pt).

The memo must address:

1. Your estimate of the causal effect of crossing the age-25 cutoff on annual healthcare spending, with a standard error (or confidence interval).

2. What this estimate means in plain language.

3. The research design behind your estimate: what method you used, why it is appropriate, what assumption it relies on, and at least one diagnostic check probing whether that assumption holds. If a diagnostic turns up a problem, explain what you do about it.

4. Threats to validity — what could still go wrong, and what population your estimate applies to.

5. A recommendation for the Ministry. The Ministry's underlying concern is about long-run engagement, not just spending at age 25. Be clear about what your estimate can and cannot tell them about the proposed policy change.

Tables, figures, and any references count toward the 2-page limit.